TRIP DATA ANALYSIS

A business analysis of casual and annual-member rider behaviour

Strategic focus
Maximizing annual memberships through evidence-led rider segmentation and marketing.

PREPARED BY	PROJECT	REPORT BASIS
Kunal	Bike-share company	Supplied presentation and embedded charts

Reporting note: The source presentation is dated October 2026 and the project filename refers to 2025. The charts show January–December but do not visibly label the year. This report therefore describes the displayed 12-month analysis and does not independently verify the source period.

1. Executive Summary

This project examines differences between casual riders and annual members to identify opportunities to increase annual memberships. Across the monthly chart, the project records **9,679,269 rides**: **6,136,573 member rides (63.4%)** and **3,542,696 casual rides (36.6%)**. Member rides form the larger share of recorded usage, while casual riders show longer average ride duration.

9,679,269	63.4%	22 min	September
Total rides in monthly chart	Member share of rides	Casual average duration	Peak month for both groups

Headline finding: the two rider groups display different weekly patterns. Member rides are stronger on weekdays, whereas casual rides are higher on a per-day average at weekends. This suggests that acquisition and retention messaging should be tailored to distinct use occasions rather than treating all riders as one segment.

The presentation proposes targeting office workers and considering limitations on casual rides. The evidence supports testing weekday commuter-oriented membership messaging, but it does not by itself establish that casual-ride restrictions would improve conversion. Such restrictions should be evaluated carefully through controlled pilots.

2. Business Context and Objective

The business problem identified in the presentation is the relationship between casual and membership riders. The stated solution is to extract meaningful insights and use them to design a marketing strategy that converts casual riders into annual members. The analytical framing is to understand the day-to-day impact on riders and identify rider needs.

Primary objective

Increase annual membership adoption by understanding when and how casual and member riders use the bike-share service, then matching membership propositions to relevant rider needs.

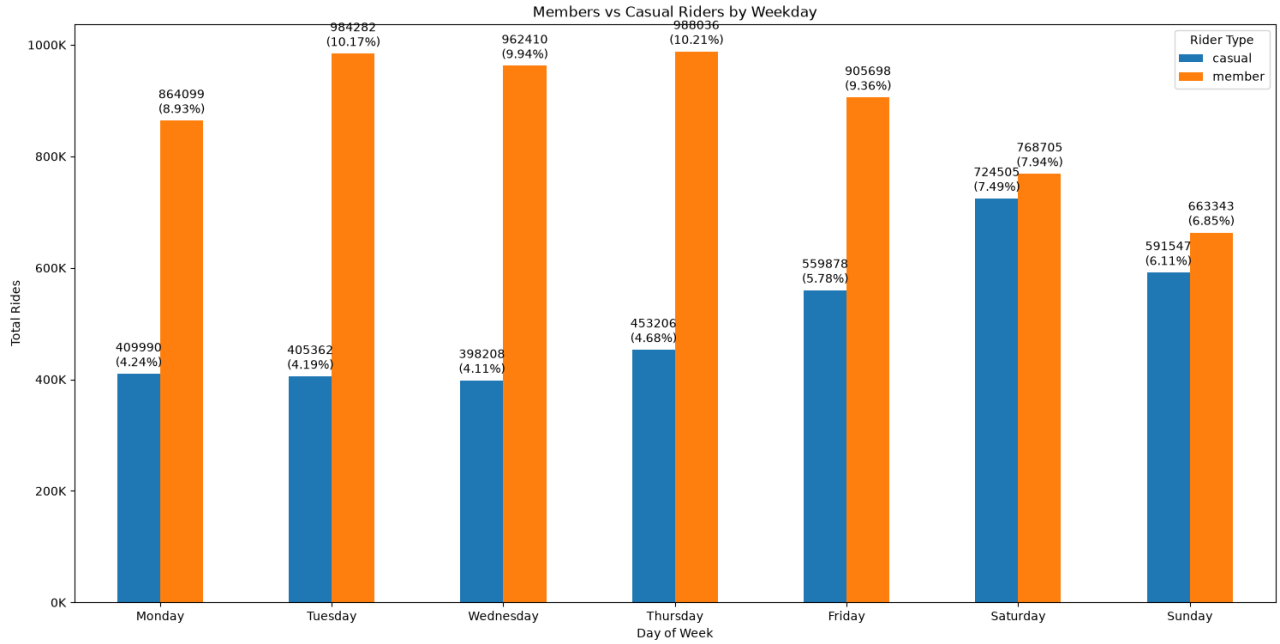
3. Scope and Analytical Approach

The supplied presentation contains three principal evidence areas: weekly rides by rider type, monthly rides by rider type, and average ride duration by rider type. The charts are used here as the source of the reported values. The presentation also identifies SQL, Python and Power BI as part of the wider project workflow in the project context shared in this conversation; the uploaded deck itself does not document detailed query logic, cleaning rules, exclusions or dashboard measures.

The report does not infer demographic profiles, trip purposes, revenue, conversion rates, customer counts or causal relationships because those measures are not present in the supplied slides.

4. Weekly Rider Behaviour

The weekday chart shows member rides exceeding casual rides from Monday through Friday. On Saturday and Sunday, casual rides exceed member rides. This is more precise than describing both groups as declining over the weekend: the weekend pattern differs by segment.



Across Monday–Friday, the average daily volume is approximately **940,905 member rides** and **445,329 casual rides**. Across Saturday–Sunday, the average daily volume is approximately **716,024 member rides** and **658,026 casual rides**.

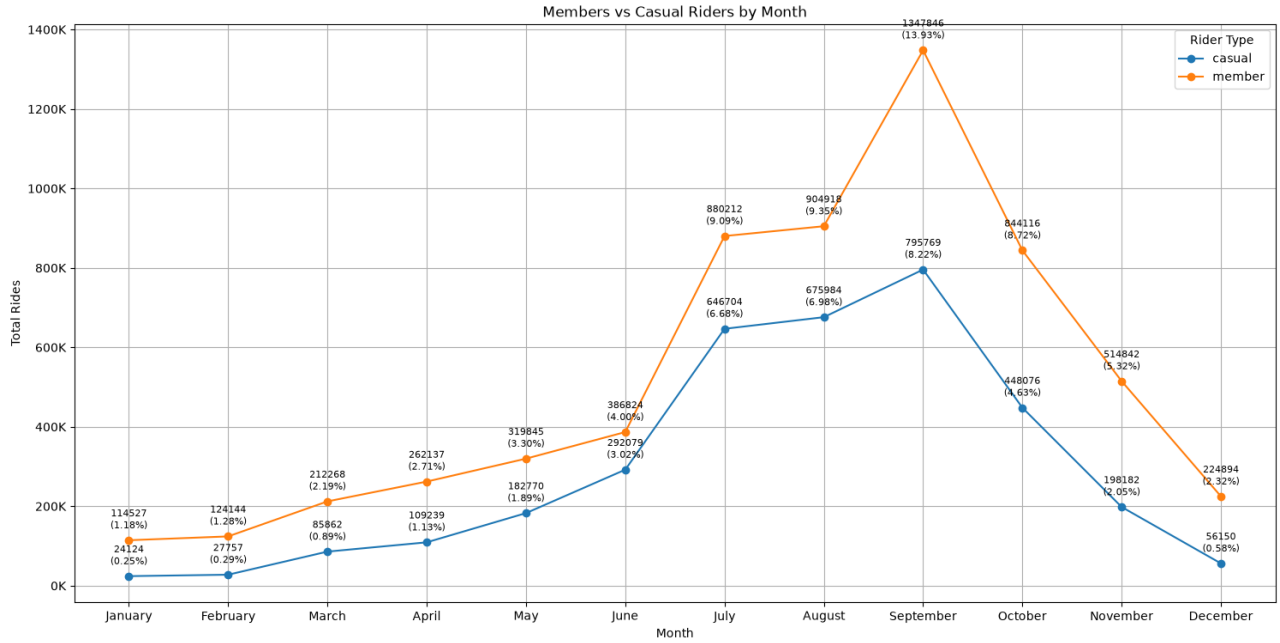
Interpretation: member usage is more concentrated on weekdays, which is consistent with a potential workday or routine-use segment. Casual usage has a stronger weekend pattern, which may reflect leisure or flexible-use occasions. The chart alone cannot prove that members are office workers or that casual riders are leisure users; trip-purpose or survey data would be needed to validate those explanations.

Business implication

Position annual membership around convenience, predictable weekday use and the cumulative value of frequent rides. For casual riders, test weekend-focused messaging that explains the benefits of membership for repeat use without assuming that every casual rider has a commuting need.

5. Monthly Trend and Seasonality

Both rider groups rise through the middle of the displayed year and reach their highest monthly ride volume in September. Member rides peak at 1,347,846 in September, while casual rides peak at 795,769 in the same month. Both groups then decline from October through December.



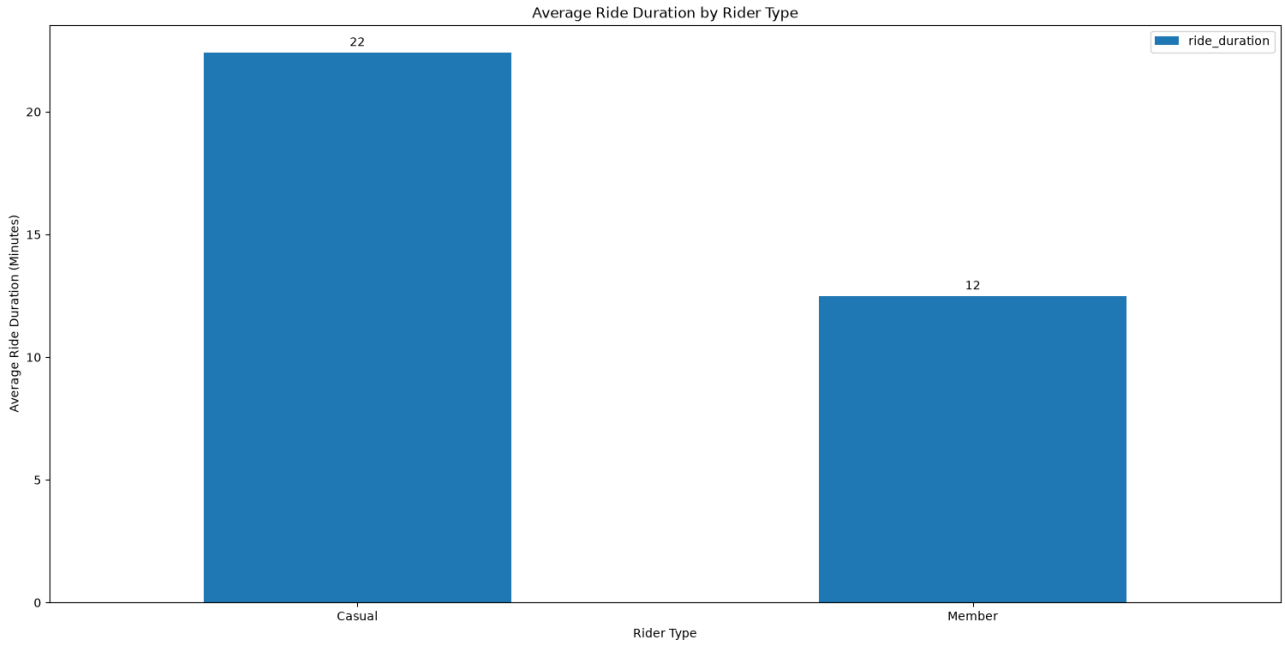
Interpretation: the shared September peak suggests a seasonal demand pattern affecting both segments. The strong increase from June to September is an opportunity to schedule membership campaigns before and during the high-activity period. The subsequent decline indicates that retention and off-season engagement may also matter.

Recommended actions

- Launch a pre-peak membership campaign before the summer-to-autumn growth period.
- Use the September peak as a high-intent period for membership upgrade prompts.
- Plan retention communications and value-led offers for the October–December decline.
- Compare the pattern with weather, holidays, service availability and year-over-year data before attributing the trend to a specific cause.

6. Average Ride Duration

The duration chart reports an average of 22 minutes for casual riders and 12 minutes for members. Casual rides are therefore 10 minutes longer on average, or approximately 83.3% longer than member rides.



Interpretation: casual riders appear to have longer average trips, while members make shorter trips on average. This difference can inform segment-specific messaging and product design. It does not establish why trips are longer, nor does it prove that restrictions would increase membership conversion.

Recommendation on casual-ride limitations

The presentation suggests adding limitations to casual rides. Before implementing any restriction, the company should assess customer impact, pricing fairness, complaints, churn and conversion. A safer first step is to test optional membership benefits, ride bundles or upgrade offers against a control group. Any policy change should be supported by measured incremental conversion and customer-experience results.

7. Strategic Recommendations

Priority	Recommendation	How to evaluate
1. Weekday conversion	Test commuter-oriented annual membership messaging for casual riders who use the service on weekdays.	Conversion rate; repeat weekday usage; membership starts.
2. Weekend acquisition	Test weekend passes or upgrade prompts for casual riders with recurring weekend activity.	Offer redemption; upgrade rate; repeat rides.
3. Seasonal campaign	Schedule campaigns ahead of the June–September growth and peak period.	Campaign lift versus a control group; cost per acquired member.
4. Retention	Use targeted communications during the October–December ride decline.	Renewal intent; active member rides; churn.
5. Product / pricing tests	Pilot optional benefits or pricing bundles before introducing broad casual-ride restrictions.	Incremental conversion, revenue, complaints and retention.

8. Measurement Framework

To determine whether the strategy is working, the business should track a small set of consistent KPIs before and after each campaign. Suggested measures include casual-to-member conversion rate, new annual memberships, repeat ride frequency by segment, member retention, average ride duration, campaign cost per acquired member and customer complaints. These measures are recommendations for future tracking; they are not reported in the supplied presentation.

Where possible, use a controlled experiment: compare eligible riders who receive a campaign with a similar group that does not. This will help separate campaign impact from seasonal changes in overall ride demand.

9. Limitations and Data Considerations

• The slides do not state the exact dataset date range or clearly label the year shown in the monthly chart.
• No total dataset row count, data dictionary, cleaning rules, missing-value treatment or outlier handling is documented in the deck.
• The duration chart shows 22 and 12 minutes, but does not specify whether the mean was calculated after removing invalid or extreme trips.
• Weekly and monthly charts report ride counts, not unique riders; ride volume should not be interpreted as customer count.
• The suggestion that members are mostly office workers is an interpretation in the presentation, not a verified demographic or trip-purpose result.
• The analysis is descriptive. It identifies patterns but does not prove that any proposed marketing action will cause membership conversion.

10. Conclusion

The analysis identifies a clear distinction between the two rider segments: member rides are stronger during the working week, casual rides have higher average daily volume on weekends, casual riders have longer average trip duration, and both groups reach their monthly peak in September. These patterns provide a practical basis for differentiated marketing.

The recommended direction is to build weekday-focused membership propositions, develop weekend-oriented conversion offers for casual riders, time campaigns around seasonal demand, and protect customer experience while testing any pricing or usage-policy changes. The next step is to validate these

hypotheses with campaign experiments and conversion KPIs.

Appendix: Key Figures Transcribed from the Presentation

Measure	Casual	Annual member
Total rides (sum of monthly chart)	3,542,696	6,136,573
Share of rides	36.6%	63.4%
Average ride duration	22 minutes	12 minutes
Peak month	September (795,769)	September (1,347,846)

Source: User-provided PowerPoint presentation, “trip_data_analysis_2025 - Copy.pptx”, including its weekly, monthly and average-duration charts. Calculations in this report are arithmetic summaries of the values displayed in those charts. No external dataset was independently reprocessed for this report.